

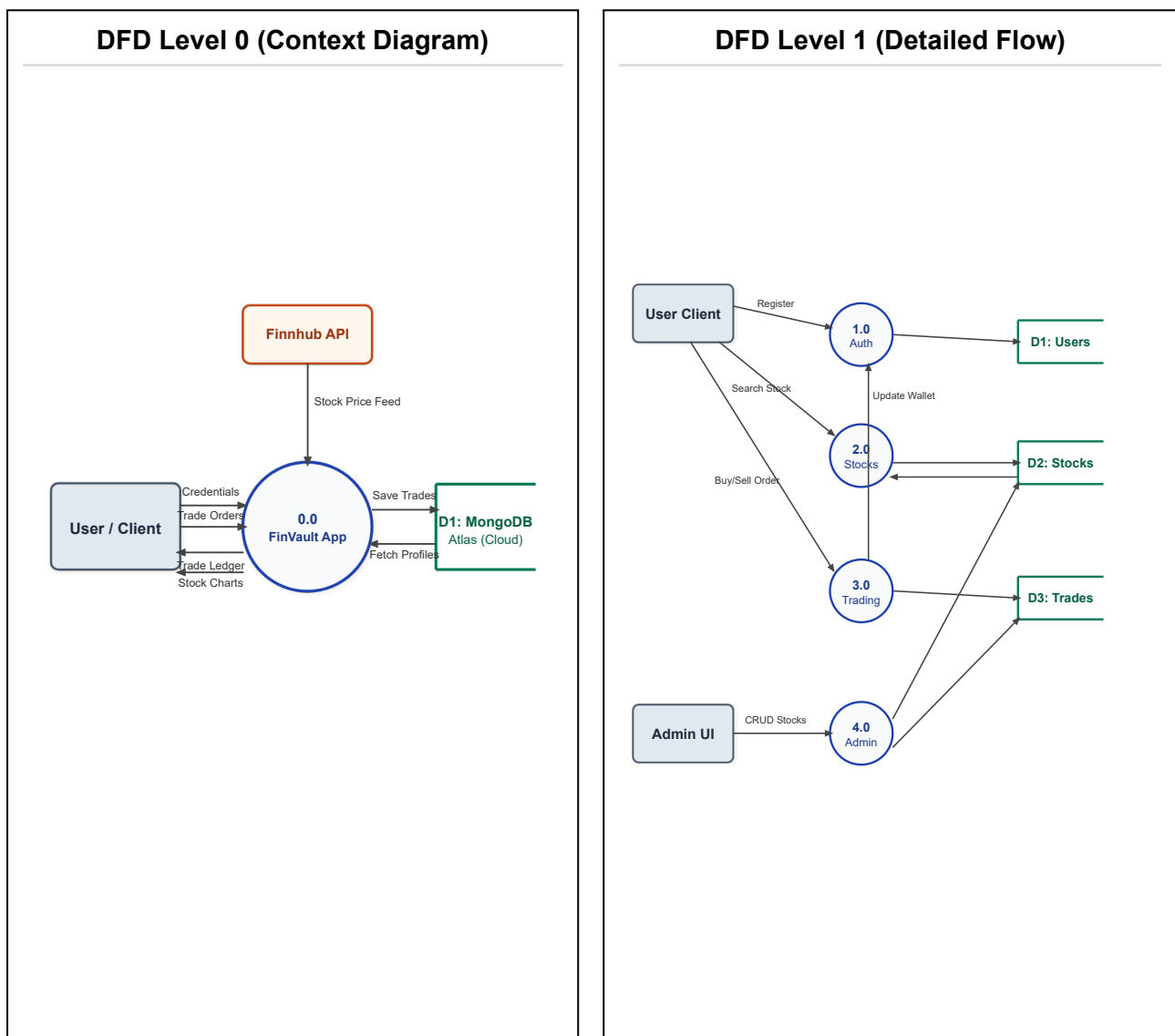
PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	T-12
Project Name	FinVault – Virtual Stock Trading Platform
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Web user)	Registration & Authentication	USN-1	As an investor, I want to register a secure account using my email and password so that I can start virtual trading.	Form validation blocks duplicate emails; passwords are hashed using bcrypt in DB.	High	Sprint-1
		USN-2	As an investor, I want to upload a custom profile picture so that I can personalize my trading dashboard card.	Multer uploads file (limit 2MB), saves path to DB, and displays avatar on profile card.	Medium	Sprint-1
	Stock Catalog & Tracking	USN-3	As an investor, I want to search listed stocks by symbol or name so that I can easily discover assets.	Listings update instantly on keypress via React search filters; show current prices.	High	Sprint-1
		USN-4	As an investor, I want to add or remove stocks from my Watchlist so that I can monitor specific assets.	Adds/removes stock ObjectId from Watchlist DB array; updates UI immediately.	High	Sprint-1
	Stock Charts	USN-5	As an investor, I want to view a historical price graph of a stock so that I can analyze past trends.	Renders line graph via Chart.js using priceHistory database records.	Medium	Sprint-2
Customer (Web user) [Cont.]	Virtual Trading System	USN-6	As an investor, I want to buy stock shares using my virtual cash balance so that I can build a mock portfolio.	Verifies wallet balance; deducts cost, inserts trade record, and updates holdings array.	High	Sprint-1
		USN-7	As an investor, I want to sell my owned stock holdings so that I can liquidate assets and secure mock gains.	Validates owned holding count; adds proceeds to wallet and updates holding/trade DB.	High	Sprint-1
		USN-8	As an investor, I want to view my transaction ledger so that I can audit my historical trading activities.	Chronological lists of BUY/SELL logs, showing price, quantity, total amount, and timestamp.	High	Sprint-1
	State Management	USN-9	As a logged-in user, I want my wallet balances and credentials to load globally so that I don't lose session context.	React Context API fetches and stores authentication and wallet balances globally.	Medium	Sprint-1
Administrator	Dashboard Overview	USN-10	As an admin, I want to view total platform users, transaction counts, and listed stocks on a single panel.	Dashboard page fetches aggregate database counts and renders KPIs on load.	Medium	Sprint-2
	Catalog CRUD Operations	USN-11	As an admin, I want to add new stocks, delete inactive listings, and edit current stock details in the catalog.	Validates admin roles; processes stock POST, PUT, and DELETE queries to the database.	High	Sprint-2
	Oversight Audits	USN-12	As an admin, I want to view a master transaction feed and list of all users to monitor system operations.	Restricted page pulls global transactions collections and user list database schemas.	Medium	Sprint-2